

Control of high-rise/high-speed elevators

Randy Roberts
Otis Elevator Company
Five Farm Springs
Farmington, CT 06034-2567
Email: robertsr@eng1.otis.utc.com

ABSTRACT:

An analytical framework for the development and evaluation of motion control system concepts for high-rise, high-speed elevators is presented in this paper. This problem definition, which includes a discussion of typical control system performance requirements, plant model dynamics and uncertainties, and control system robustness requirements, serves as a benchmark for the application and evaluation of advanced control approaches. Unique features of this problem include time varying dynamics, uncertain structural response, noncolocated control, sensor suite selection, and separable command and disturbance rejection control system requirements.

1. Introduction

Maintaining or improving elevator performance as building rises and car speeds increase will require new and advanced control system technologies. Just the 450 meter (1480 ft.) Petronas Towers in Malaysia near completion which will make them the world's tallest buildings, work has begun on the 460 meter (1510 ft.) Shanghai World Financial Center in China [1]. This upward trend is likely to continue as a number of proposed plans have been announced for mega high-rise buildings to be constructed in the next millennium with rises in excess of 500 meters, some even topping 2000 meters [2]. As might be expected, there are a number of major engineering challenges associated with the design of such mega structures, including construction methods, structural design, infrastructure design, and building transportation (elevator) system design [3].

A number of issues arise in the design of effective transportation systems for tall buildings using conventional roped elevator technology [4]. These design problems can be grouped into three general areas; efficiency, passenger comfort, and safety.

Efficiency can be measured in many ways, but ultimately must be related in some way to the following:

- how quickly can people can be moved into and out of a building or buildings,
- what level of energy and power is required to achieve these transitions, and
- how much volume in the building is consumed by the candidate transportation system.

One way to address this efficiency issue is to increase the maximum speed of high-rise roped elevators from the current top end of 750 meters/minute up to 1000 meters/minute or higher in the future [5]. In addition, a

number of concepts have been proposed in which multiple elevator cabs can operate in the same hoistway using traditional roped elevator propulsion [6] or ropeless elevator propulsion (e.g., on car linear induction or synchronous motors) [7].

Another clear goal and priority of these ultra high-rise elevator systems is that they create a comfortable environment for their passengers who after all are occupants or visitors to these monumental landmark buildings. Three metrics which are commonly defined to assess passenger comfort in elevators are cab vertical and horizontal vibration levels, noise, and ear comfort. Acceptable levels of vertical and lateral vibration levels in these high speed elevators are very small and are typically expressed in milliG target levels (e.g., 8-10 milliGs).

The design of a control system to regulate the motion of a roped elevator is made difficult due in part to the inherent noncolocated control problem posed by this application. That is, there exists finite transmission delays in the potentially very long hoistway ropes between the actuation source (i.e., the drive motor at the top of the building) and the elevator car which result in significant phase delays which must be considered in the control design.

The performance requirements for elevators are very stringent, in a large part due to their live "cargo". Present positional vertical accuracy requirements for roped elevators are typically in the range of 0.001% to 0.005% of the full range of motion in high-rise elevator applications (e.g., 6 millimeters over potential rises up to 500 meters). Another complicating factor is the natural shift in the low frequency hoistway resonant modes to lower frequencies as the rise increases, thus it becomes more difficult to achieve a stable well damped response at the elevator car in response to commanded positional changes.

Another consequence of increases in building rises is the associated decrease in effective rope stiffness which results in a requirement for higher position system feedback bandwidth to minimize cab to building sill gaps during passenger loading and unloading transients.

In this paper a framework will be developed for the analytical assessment and development of advanced motion control concepts for a hypothetical 500 meter rise roped elevator system. This problem definition is similar to other robust control benchmark problems [8], but also includes the following:

- the developed problem is representative of a real world application area with supplied models being validated using experimentally collected field measurements,

- time varying linear dynamics are included in the system plant model to exercise various options for control system design, and
- the control problem is broadened to include sensor selection and definition.

2. Elevator Hoistway Dynamics Model

The basic roped elevator dynamic system, illustrated in Figure 1, consists of four major inertial elements; the drive sheave, elevator car, counterweight, and compensation sheave.

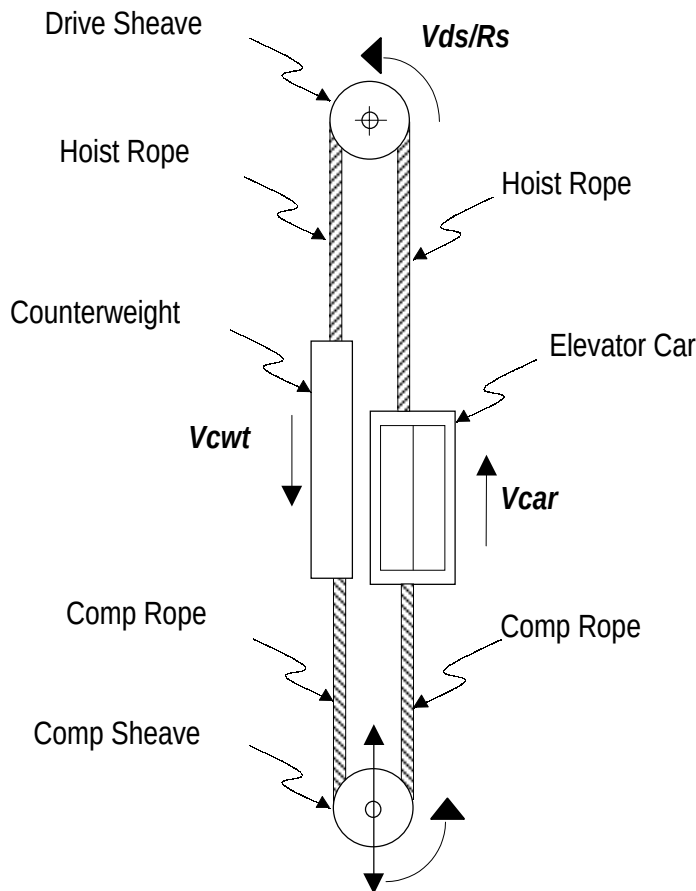


Figure 1: Elevator Physical Configuration

There are five major degrees-of-freedom (DOF) in this configuration; rotation of the drive and compensation sheave, and translation of the car, counterweight, and compensation sheave. Each of the major inertial elements are connected together with rope segments of whose lengths vary as the elevator car moves up and down the hoistway.

The control input is the motor torque applied at the drive sheave and the fundamental controlled variable is the vertical position of the elevator cab.

A model of this physical system is presented in Figure 2.

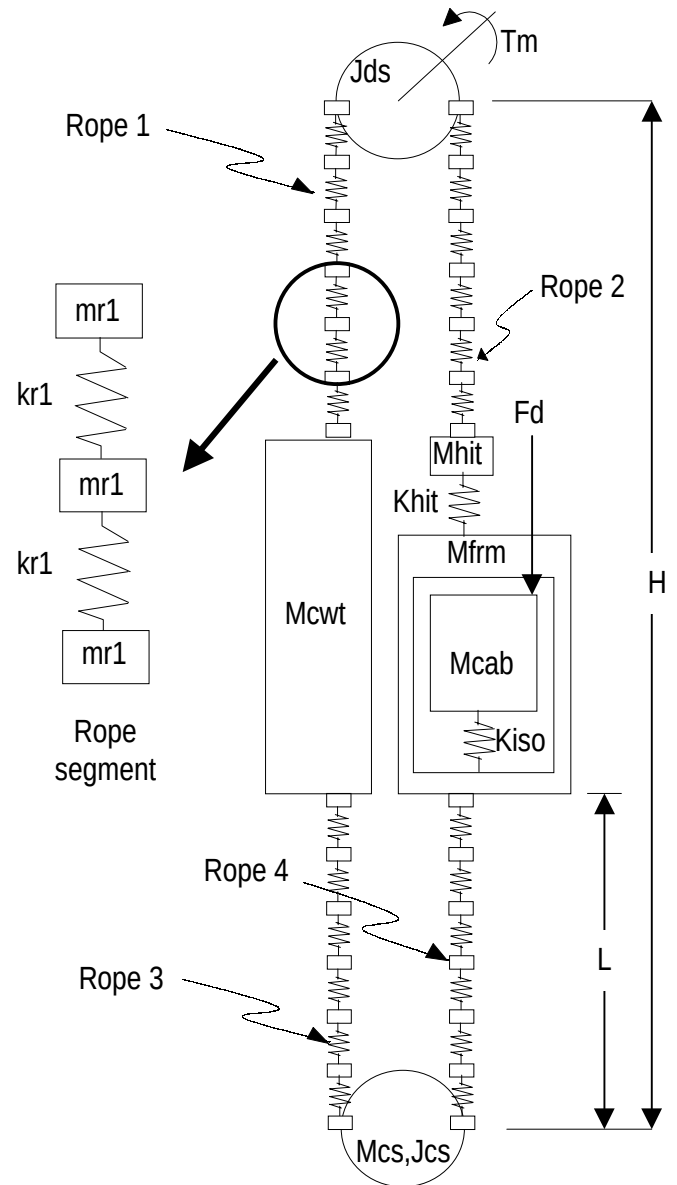


Figure 2: Elevator Dynamical Model

A 3 DOF model of the elevator car is assumed representing the elevator frame, cab, and rope hitch. The parameter values for this car model are fixed with the exception of the cab mass (M_{cab}) which is a variable quantity dependent on the number of elevator passengers. This relationship is given by:

$$M_{cab} = M_{cabo} + p \cdot \text{Duty}$$

where M_{cabo} is the empty cab mass, Duty is the maximum allowable passenger load, and p is a value from 0-1 representing the degree of loading.

Another critical feature of this model is the representation of the structural dynamics of the hoistway ropes themselves using a lumped mass approximation to capture the finite force transmission delays in each of the four different rope segments. Each of these rope segment

models represents the aggregate response of what is actually a set of multiple (e.g., 5-9) ropes.

The properties of the hoist ropes (i.e., rope 1 and rope 2 which connect the car and counterweight and which ride on the drive sheave) are:

- M_{dr} = rope density (mass/unit length)
- K_{dr} = rope stiffness/unit length, and
- C_{dr} = rope damping/unit length.

Similarly, properties of the compensation ropes (i.e., rope 3 and rope 4 which are at the bottom of the car which ride on the compensation sheave) are:

- M_{cr} = rope density (mass/unit length)
- K_{cr} = rope stiffness/unit length, and
- C_{cr} = rope damping/unit length.

The parameter values for any lumped rope section model are determined by its rope type (i.e., hoist or compensation) and length. The values for the lumped segment mass and stiffness parameters for these four rope segments can be approximated as:

$$\begin{aligned} m_{r1} &= M_{dr} * L / (n+2) & k_{r1} &= n * K_{dr} / L \\ m_{r2} &= M_{dr} * (H-L) / (n+2) & k_{r2} &= n * K_{dr} / (H-L) \\ m_{r3} &= M_{cr} * (H-L) / (n+2) & k_{r3} &= n * K_{cr} / (H-L) \\ m_{r4} &= M_{cr} * L / (n+2) & k_{r4} &= n * K_{cr} / L \end{aligned}$$

where $n=5$ (i.e., a five dynamic lumped mass approximation) is used.

Not shown in Figure 2, but included in the dynamic model representation, are viscous friction dampers in parallel with each spring. For the lumped rope models these terms are represented in a similar manner to the stiffness terms as:

$$\begin{aligned} c_{r1} &= n * C_{dr} / L & c_{r2} &= n * C_{dr} / (H-L) \\ c_{r3} &= n * C_{cr} / (H-L) & c_{r4} &= n * C_{cr} / L \end{aligned}$$

Relative damping values in parallel with cab isolation pads (Kiso) and rope hitch (Khit) are Ciso and Chit respectively. In addition, there are absolute viscous friction dampers (i.e., connected to an inertial ground) for the drive sheave (Cds), compensation sheave rotation (Ccs) and translation (Cmm), counterweight (Ccw), and elevator cab (Ccar) which are not shown in Figure 2.

Thus, the elevator hoistway dynamics can be represented as a time varying linear state space model of the form:

$$\begin{aligned} dx/dt &= A(L,p)x + B(L,p)u \\ y &= C(L,p)x + D(L,p)u \end{aligned}$$

where p is the payload loading fraction and L is the instantaneous vertical position of the elevator. In general, p is a fixed constant during elevator transitions and L is a time varying quantity. For design purposes one can define the local linear hoistway dynamics given a floor position (Flr) by setting

$$L = D_o + D_{flr} * Flr$$

where D_o is the nominal spacing at the bottom of the hoistway, D_{flr} is the floor to floor spacing, and Flr is the

floor landing (i.e., 1-135 in this fictitious 500 meter building).

The input vector to the plant model are;

- $u(1)$ = drive sheave motor torque (T_m), and
- $u(2)$ = a disturbance force (F_d) which acts on the elevator cab.

There are 27 DOFs in this model, thus the state vector (x) is a 54x1 vector. Theoretically any DOF associated with the major system inertial elements is measurable and therefore a candidate for control. However, the main control measurements are the drive sheave velocity (V_{ds}) and the car position (P_{car}).

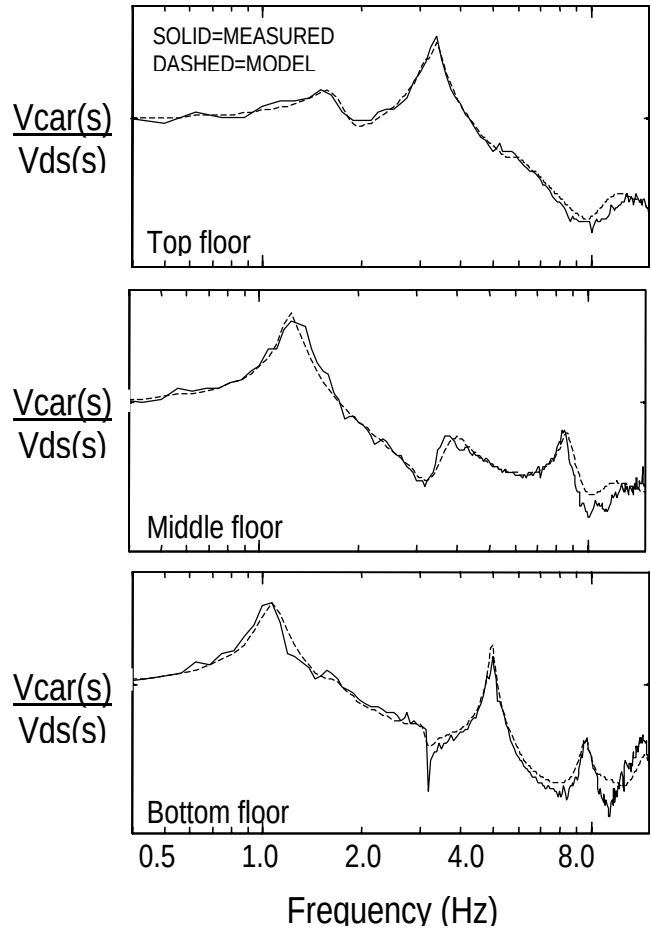


Figure 3: Model Validation Results

This dynamic model has been validated using experimentally derived transfer function data collected at a high rise elevator installation. Figure 3 shows a plot of the transfer function from V_{ds} to the car velocity (dP_{car}/dt) as measured in three locations (top, midrise, and bottom) of a high rise elevator. In this case, $p=0.5$, and L was fixed at three values. Small signal excitation commands were injected into the system and V_{car} and V_{ds} were measured. It can be seen that this analytical model closely replicates the measured response, capturing the downward shifting of resonant modes as the car to drive sheave distance increases.

3. Controller Performance Assessment Metrics

The performance objectives of the elevator vertical motion control system which are considered here are:

1. to move the elevator from one vertical position (i.e., floor) to another in a controlled manner (i.e., minimize flight times subject to constraints), and
2. to minimize the amount of car to sill dynamic deflection during passenger loading and unloading transients.

These requirements, in addition to others, will be explicitly defined in this section.

Criteria 1) One floor transition response requirements:

The control system must move the cab position from vertical position A to position B in such a manner as to ensure minimum cab vibration levels while landing in the shortest time within certain guidelines for landing accuracy and overshoot. For purposes of this analytical assessment, a time-based command trajectory will be assumed to move the car over a distance of 3.66 meters (12 ft.).

In this case, a two step process is used to create the command trajectory. First, a minimum time trajectory is calculated subject to limits of dictated acceleration (1 m/s^2) and jerk (1.5 m/s^3). Next, this trajectory is passed through a 1st order low pass filter with a time constant of 0.5 seconds. Figure 4 presents plots of the resultant acceleration, velocity, and distance to go as a function of time. It can be seen for this constrained case the ideal flight time (i.e., the time to within 6 mm of the destination) is 6.2 seconds.

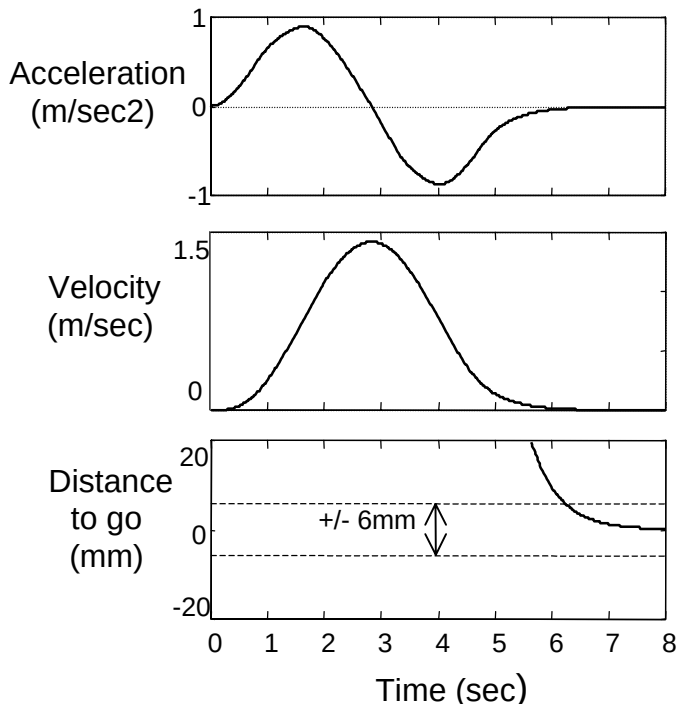


Figure 4: Dictated Trajectories

The measure of the one floor flight time response will be the flight time, landing accuracy, and overshoot (i.e., the

maximum incremental displacement beyond the final settling position).

Criteria 2) Releveling transient response requirements:

A second function of the elevator motion control system is to maintain the cab to sill gap during periods of passenger loading and unloading. This is a disturbance rejection property of the control system which can be modeled by injecting a time dependent disturbance force (F_d) into the simulation.

For analytical assessments the assumed time history of F_d is as shown in Figure 5 below. This assumes that the rate of passenger loading is constant and corresponds to a 15 second period to reach the maximum duty level (1800 kg in this case). This profile is smoothed using a 1.0 second 1st order lag filter.

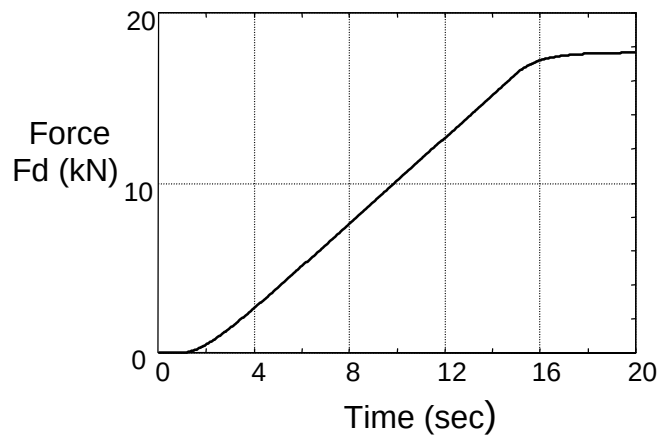


Figure 5: Assumed Disturbance Profile

It is desired to minimize the cab displacement during this transient. A guideline is that the maximum value of this displacement should be less than 6 mm.

Criteria 3) Control system robustness: A key consideration in designing the elevator vertical motion control system is that it should be robust to variations in modeled dynamics and unmodeled dynamics. It is well known that the stiffness of typical elevator hoistway ropes can vary considerably during its lifetime due to its construction geometry and materials. Thus, expected ranges of parameter values (e.g., stiffness, damping, and mass) are defined over which any developed control system must achieve robust performance.

In particular, it is desired that the control performance goals as listed above are achieved in the presence of variations in the following system model parameters:

- $\pm 20\%$ variation in the hoist and compensation rope stiffnesses (i.e., K_{dr} and K_{cr} values),
- $\pm 50\%$ variation in all damping values (i.e., C_{dr} , C_{cr} , C_{cwt} , C_{car} , C_{ds} , C_{hit} , C_{iso} , C_{cs} , C_{mm}), and
- $\pm 10\%$ variation in the torque generator gain value (i.e., $T_m = K_{fac} \cdot T_c$, where T_c is the output of the motion control system and K_{fac} ranges from 0.9-1.1).

4. Conclusions

This paper has presented an analytical framework for the design of advanced motion control concepts for high-rise, high-speed elevators. This formulation serves as a benchmark problem definition for the design and evaluation of advanced control design methodologies, including LMI-based control, model-based fuzzy control, nonlinear control, and robust control formulations.

The assessment of these control concepts involves consideration of the following issues:

- What is the command tracking performance (i.e., flight time, position accuracy, position overshoot, ride quality, etc.) and how is it affected by anticipated model variations?
- What is the releveling performance (i.e., maximum position excursion and ride quality) and how is it affected by anticipated model variations?
- What are the computational requirements for the proposed control concept?
- What sensors are used in the control concept and on what basis were they selected?

Areas for future consideration, but not explicitly part of this assessment, include:

- How can the control concept be tuned in the field? That is, what types of experimentally derived models are required to support the controller gain calculations?
- What is the optimal trajectory planning algorithm and how should its design be coordinated with the motion control system design?

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APPENDIX

Model Parameter Values

Representative values for the model parameters for the elevator dynamic model are as follows:

Inertias:

Mcabo = 3400 kg	Mfrm = 3410 kg
Mdr = 17.4 kg / m	Mcr = 10.8 kg / m
Mcs = 2470 kg	Mhit = 90 kg
Jcs = 334 kgm ²	Jds = 1622 kgm ²
Duty = 1800 kg	p = 0.5

Stiffnesses:

Kdr = 1.74e8 N	Kcr = 9.63e7 N
Kiso = 6e6 N / m	Khit = 1.3e6 N / m

Damping:

Cdr = 7.2e5 Ns	Ccr = 2.8e5 Ns
Ciso = 1.2e5 Ns	Cds = 100 kgm ² / s
Ccs = 44 kgm ² / s	Ccar = 40 Ns / m
Ccwt = 40 Ns / m	Chit = 1.3e3 Ns / m
Cmm = 2e4 Ns / m	

Dimensions:

Rs = 0.6 m	Dos = 3.0 m
H = 500 m	Dflr = 3.66 m

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